

An Opportunity for Healthier Public Housing

The Boston Housing Authority is working to make public housing more energy efficient and healthier for the occupants at the same time.

by **Kimberly Vermeer**

Public housing, much of it built more than 40 years ago, can have serious environmental problems as a result of age and deferred maintenance. Residents experience asthma and other respiratory illnesses at many times the national average, and report chronic health problems such as coughing, fatigue, colds, headaches, and allergies. In some cases, poorly functioning HVAC systems contribute to these health problems. In other cases, earlier energy conservation efforts have aggravated these conditions by reducing indoor air quality (IAQ), increasing moisture levels, and providing conditions that promote the growth of mold and pests.

But there are problems when addressing health in energy conservation plans. One is that some actions to improve health don't save energy. Adding ventilation to a previously unvented apartment, for example, won't pay for itself out of energy savings. Conversely, some standard approaches to energy conservation could aggravate health concerns. Tightening up a unit with an unvented gas range will increase indoor air pollution. And measuring success is harder when health is a factor. It is one thing to measure the temperature of an apartment after conservation measures and quite another to measure the healthiness of the apartment.

Still, some public housing authorities (PHAs) are exploring new ways to tackle combining energy and health-related retrofits.

The Standard Way

Over the last decade, some PHAs and contractors have worked out approaches to both save energy and meet the economic expectations of contractors, given the payback period



Boston's West Broadway apartments underwent an energy retrofit in 2000. Now the Healthy Public Housing Initiative is measuring the effect of the retrofit measures on the indoor environment and the health of the occupants.

limits (see "Spinning Gold"). The recent energy and water upgrades at the West Broadway development, a 540-unit property of the Boston Housing Authority (BHA), is an example of such a standard approach (see "Public Housing Breaks the Mold," *HE* Sept/Oct '01, p. 24).

West Broadway is a typical northeastern public housing development, with many three-story brick buildings built in 1939. Between 1987 and 1990, significant renovations were completed, including new windows, exterior cosmetic improvements, and interior gut rehabs that included installation of 1-inch beadboard insulation and new Sheetrock walls. By the late 1990s, however, the central steam heating and domestic hot water (DHW) system at West Broadway was not functioning well. In response, a substantial energy performance contract was performed by NORESKO, a Framingham, Massachu-

setts, energy services company (ESCO) in 2000. The contract included converting the central steam-heating and DHW system to a distributed forced hot water heating system and a DHW system with dedicated boilers, DHW storage tanks, and boiler controls for each of the 23 occupied buildings. The scope of work also included manual on-off valves for each radiator for better resident temperature control in individual apartments. A water conservation package of low-flow showerheads, low-volume toilets, and sink aerators completed the program.

But were these upgrades enough on their own to address the environmental health concerns of residents? As the connections among energy, building systems, and health become clearer, affordable housing stakeholders and some industry practitioners are beginning to realize the benefits of connecting energy and environmental health in energy conservation programs.

The Boston Approach: A Model?

With the assistance of John Snell, a consultant with Peregrine Energy, BHA is taking the innovative approach of combining the energy conservation incentives provided by HUD with capital grant funds to complete comprehensive building and system upgrades. These upgrades will address health-related building conditions as well as energy and water conservation. In so doing, BHA is attempting to leverage the energy and water conservation incentives to accomplish whole building renovations that will result in more efficient, more comfortable buildings that are also healthier homes for their residents.

In fact, BHA has made resident health and safety a key framework for capital planning and operations. "By focusing on health we can establish priorities for action with residents around shared goals," says BHA's administrator Sandra Henriquez. In addition to focusing on health in the energy and water conservation plan, BHA is a lead partner in the Healthy Public Housing Initiative (HPHI), a multiyear HUD-funded research project to investigate the health improvements that public housing resi-

dents experience from investments in building and unit interventions.

In response to the growing awareness about the link between energy systems and resident health, BHA included several health and comfort questions in the post-conversion resident satisfaction survey for West Broadway. First-year assessments indicate that savings are being achieved and that resident comfort has improved. Interestingly, fatigue, a chronic health complaint prior to the upgrades, has been reduced. HPHI is conducting a further systematic review of the benefits achieved from the energy upgrades that will compare before-and-after surveys of environmental health and comfort with environmental testing results. That analysis should be complete in early 2003. As the preliminary assessment of West Broadway indicates, some health benefits can be achieved simply by improving system performance. But much greater health benefits could be achieved if health-directed improvements were integrated into the energy upgrade program.

For 58 developments with over 11,000 units, BHA is moving forward with a blended approach of performance contracts and financing, with additional subsidy agreements that will enable it to

incorporate far more actions targeted to resident health and comfort. Grants from Keyspan Energy Services and Nstar, BHA's gas and electric utility providers, are paying for feasibility studies, engineering, and advanced technology pilot projects. These grants are in addition to the technical and financial support Keystone and Nstar provide for standard energy efficiency programs. Improvement measures will include air sealing (for energy savings and pest control), water leak repairs (for moisture and pest control), new or upgraded mechanical ventilation (for improved air quality and moisture control), and apartment radiator controls for greater thermal comfort.

At the Franklin Hill Development, for example, BHA's plans are well under way for integrating utility company conservation incentives with existing capital grant funds and HUD energy project funding mechanisms. This fall and winter, BHA will pilot-test a steam-heating system retrofit/commissioning approach with a utility grant while awaiting HUD approval for a complete overhaul using HUD incentives in 2003. Basement moisture control upgrades—also expected to reduce pest problems—are in the design stage. Work

Spinning Gold

In the United States, federally funded PHAs receive formula-based annual operating subsidies from HUD to cover the difference between resident payments of 30% of income and actual operations costs, including utilities. During the energy crisis of the 1970s, policymakers realized that the full reimbursement of utilities gave PHAs no incentive to try to reduce energy costs. Furthermore, the subsidy system provided no funding mechanism to pay for the up-front cost of conservation measures. In response, Congress passed legislation to enable HUD and PHAs to share utility cost savings as an incentive to encourage conservation. The original legislation allowed for a 4-year payback period. However, this provided too little time to make effective

changes, and little conservation work occurred. In 1991 the legislation was amended to increase the payback period to 12 years. It took HUD another 2 years to develop the regulations. The Green Book, issued in 1994, gave PHAs four incentive mechanisms for implementing energy conservation.

Savings retention. PHAs make direct investments of operating and/or capital funds. They can retain 75% of energy and water savings for three years to refund the cost of short-term payback or low-cost measures, such as replacing water heaters, repairing water leaks, or installing compact fluorescent light fixtures.

Utility rate reduction. PHAs that negotiate rate reductions with utility providers can keep 50% of the resulting savings.

Frozen base. PHAs can contract with ESCOs for conservation measures and retain 100% of the savings for up

to 12 years to pay off the financing of up-front costs.

Additional subsidy. Financing can be supported by additional operating subsidy agreements for a term negotiated with HUD.

In practice, frozen-base incentives using energy performance contracts have been the most-used mechanism to date. As a result of the 1994 regulations, a number of ESCOs have built up their businesses by providing energy performance contracts to PHAs. As of 2001, PHAs nationwide had implemented 87 incentive plans, saving \$31.9 million in energy and water costs annually. Another 42 proposals are in the review process. Energy performance contracts have been by far the most popular approach, used in 51 of the 87 completed plans.

will be completed in late 2002 or early 2003, using capital grant funds. Roof replacement with a special focus on moisture control—another capital grant project—is in design, with work scheduled for early 2003.

As part of HPHI, BHA will install apartment ventilation in 25 to 30 apartments, and the Initiative will track health and environmental changes. Later plans may include ventilation for all apartments. Water conservation—low-volume toilets, low-flow showerheads, and sink aerators—will be bundled with the steam system overhaul in 2003.

Will Others Go for It?

Will other housing authorities take advantage of the opportunities to address health issues through energy conservation? John Snell is confident that as knowledge and experience of both housing authorities and energy performance contractors increases, they will. However, since performance contractors are likely to continue as the primary implementers of energy and health strategies, their concerns must be addressed.

Vicky Hayes, a consultant who works with Siemens Building Technologies, believes that ESCOs can and should lead the way. Incentives to do so include reducing risks, increasing the scope of work on a given project, and expanding the set of services ESCOs can offer. She points to the growing awareness of moisture and mold problems as a potential liability factor that ESCOs should start factoring into their calculations. In her thinking, ESCOs should begin to take responsibility for addressing these larger environmental health issues. “I get excited about the possibilities, because these are problems ESCOs can solve, provided they bring the right expertise to the team.”

Some industry experts believe that ESCOs use very conservative savings estimates to protect themselves against the savings guarantees, and that PHAs could insist on a larger scope of energy- and health-related work without affecting the willingness of ESCOs to submit a bid. This may happen as PHAs learn more about the benefits to be gained from combining energy conservation

with health-related improvements. In the meantime, Matt Pesce, an energy and housing consultant for the DOE Rebuild America program, recommends that PHAs utilize the additional subsidy agreement mechanism (financing that is supported by additional operating subsidy agreements with HUD), at least for the health-related part of the work. Used alone or in combination with the ESCO approach, the additional subsidy agreement method greatly increases flexibility for PHAs and enables them to incorporate more elements into their conservation programs.

Encouraging Attention to Health

The compelling need for PHAs to find innovative strategies to address the backlog of capital needs will push the industry toward a more comprehensive approach to energy conservation over the next several years, but other policy changes that will encourage better integration of energy and health are in the works.

First, HUD has developed a new energy strategy to address energy conservation across its portfolio of housing, including its Office of Public and Indian Housing as well as the FHA-insured portfolio. The action plan, implemented in late 2002, calls for improvements to the public housing conservation incentives programs, and for setting higher performance benchmarks. The strategy also calls for HUD's Office of Healthy Homes and Lead Hazard Control to work with other HUD divisions, EPA, and DOE to develop programs and protocols for energy-efficient and healthy housing, especially in weatherization and rehabilitation programs.

Second, the Senate energy bill (S517) includes an amendment to increase energy efficiency in public housing in a number of ways. One element that would increase the ability of PHAs and ESCOs to be more creative would extend the years of shared savings in the frozen-base incentive (PHAs contract with ESCOs for conservation work and then pay for the work from the energy savings) from 12 years to 20 years. This will allow for more innovative approaches to saving energy but will also give PHAs and ESCOs room to

incorporate health actions into their plans. Matt Pesce reports that the public housing energy efficiency amendment has cleared the conference process and at press time awaits passage of the bill.

Third, HUD and the housing industry have “discovered” mold as a serious housing and health issue. Some industry people feel that mold will soon be getting the attention that asbestos and lead got in the past, and that HUD subsidy and financing programs will be modified to address the environmental factors that contribute to mold problems.

With housing authorities looking for creative solutions to their capital needs, they will be watching to see how well Boston succeeds in meeting its ambitious goals, and whether HUD's growing awareness of the link between energy and health encourages housing authorities to adopt this innovative approach. If so, the growing partnerships between PHAs and ESCOs for energy conservation can expand to become partnerships to create healthier home environments for some of our nation's most vulnerable citizens.



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